Task 5: Transaction-Based Payment Simulation

Aim: To simulate an online payment system using database transactions by deducting the amount from a user account and adding it to a merchant account, ensuring data consistency using COMMIT and ROLLBACK operations.

Algorithm:

1. Start the program
2. Enter payment amount
3. Begin transaction
4. Check user account balance
5. If balance is sufficient

- Deduct amount from user
- Add amount to merchant

6. If all operations successful → COMMIT
7. If any error occurs → ROLLBACK
8. Display success or failure message
9. Stop the program

Program:

db.php:

```
<?php

$conn = new mysqli("localhost", "root", "", "bank_db");

if ($conn->connect_error) {

    die("Connection failed: " . $conn->connect_error);

}

?>
```

Payment.php:

```
<?php

include 'db.php';

if ($_SERVER["REQUEST_METHOD"] == "POST") {

    $amount = $_POST['amount'];

    // Start Transaction

    $conn->begin_transaction();
```

```
    try {
        // Get User Balance
        $result = $conn->query("SELECT balance FROM accounts WHERE name='User'");
        $row = $result->fetch_assoc();
        $user_balance = $row['balance'];
        // Check Balance
        if ($user_balance < $amount) {
            throw new Exception("Insufficient Balance!");
        }
        // Deduct from User
        $conn->query("UPDATE accounts SET balance = balance - $amount WHERE
name='User'");
        // Add to Merchant
        $conn->query("UPDATE accounts SET balance = balance + $amount WHERE
name='Merchant'");
        // Commit Transaction
        $conn->commit();
        $message = "Payment Successful!";
    } catch (Exception $e) {
        // Rollback Transaction
        $conn->rollback();
        $message = "Transaction Failed: " . $e->getMessage();
    }
}
// Fetch Updated Balances
$result = $conn->query("SELECT * FROM accounts");
?>
<!DOCTYPE html>
<html>
<head>
```

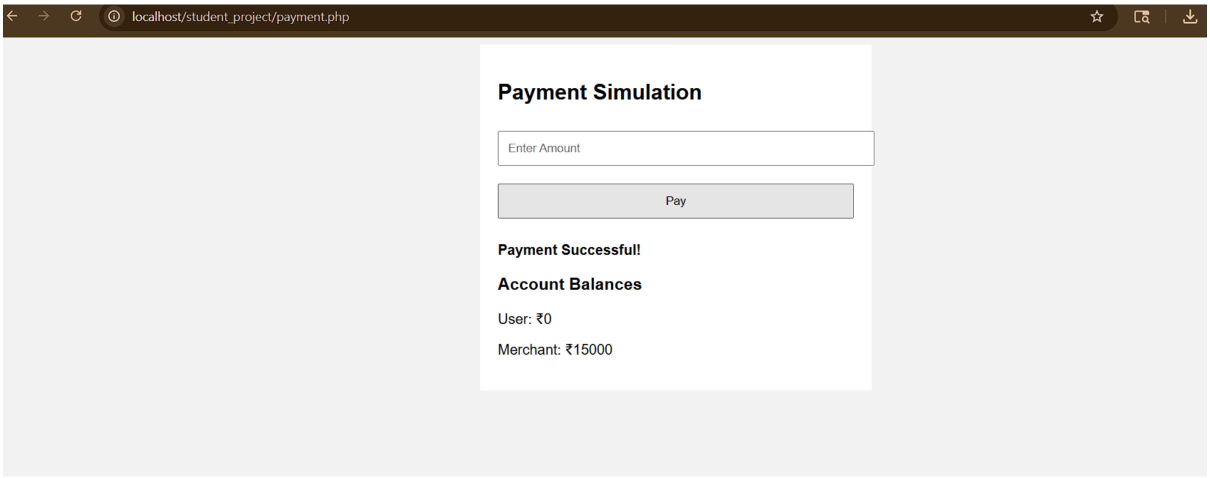
```
    <title>Payment System</title>
    <style>
        body { font-family: Arial; background: #f2f2f2; }
        .box {
            width: 350px;
            margin: 100px auto;
            background: white;
            padding: 20px;
        }
        input, button {
            width: 100%;
            padding: 10px;
            margin: 10px 0;
        }
        .msg {
            font-weight: bold;
            color: blue;
        }
    </style>
</head>
<body>
<div class="box">
    <h2>Payment Simulation</h2>
    <form method="POST">
        <input type="number" name="amount" placeholder="Enter Amount" required>
        <button type="submit">Pay</button>
    </form>
    <?php if(isset($message)) echo "<p class='msg'>$message</p>"; ?>
```

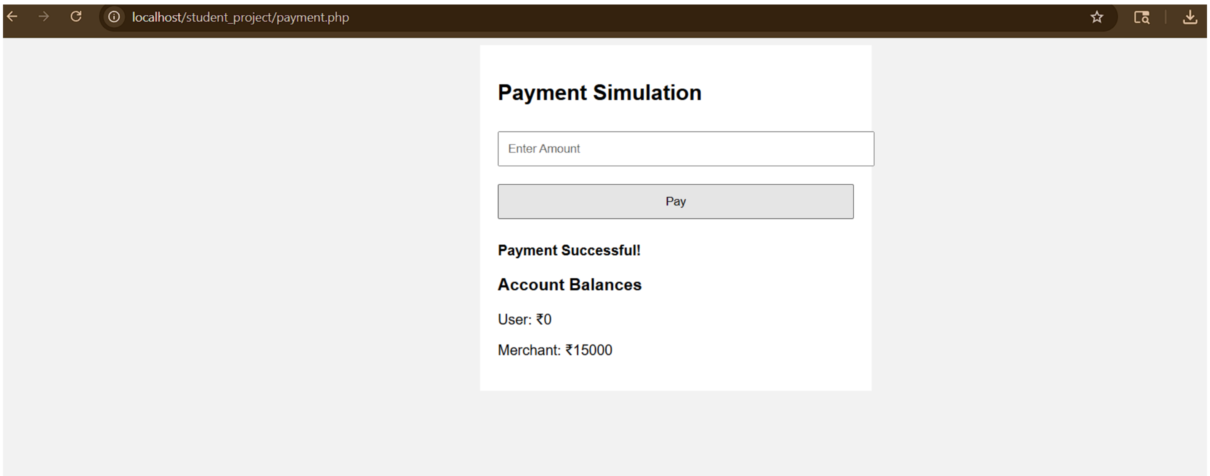
```
    <h3>Account Balances</h3>
    <?php while($row = $result->fetch_assoc()) { ?>
        <p><?php echo $row['name']; ?>: ₹<?php echo $row['balance']; ?></p>
    <?php } ?>
</div>
</body>
</html>
```

Database setup:

```
CREATE DATABASE bank_db;
USE bank_db;
CREATE TABLE accounts (
    id INT AUTO_INCREMENT PRIMARY KEY,
    name VARCHAR(50),
    balance INT
);
INSERT INTO accounts (name, balance) VALUES
('User', 10000),
('Merchant', 5000);
```

Output:





Result: Thus, the execution of the transaction based payment simulation has been successfully executed and output generated .